

Measure and Integration for Probability

Topic 1

What this topic builds

By the end of this topic, you should be able to:

- ① understand why probability is defined on a sigma-algebra of events;
- ② understand the Lebesgue integral;
- ③ understand and manage monotone convergence theorem, dominated convergence theorem, Fatou's lemma.

Contents

Consider a probability space $(\Omega, \mathcal{F}, P)$

$\downarrow$ sample space $\downarrow$ σ -algebra $\searrow$ probability measure.

1. Probability spaces and measurable events
2. From Jordan content to outer measure
3. Lebesgue measurable sets and measure
4. Lebesgue integration
5. MCT, Fatou's Lemma, DCT, BCT

Where we are

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A probabilistic model starts with complete possible worlds

Definition 1 (Sample point and sample space) A probabilistic experiment is represented by a set Ω . Exactly one element $\omega \in \Omega$ is realized. The element ω is a sample point, and Ω is the sample space.

random
variable
 $\uparrow$

Tossing 1 coin

$\Omega = \{H, T\}$, one realized outcome, $\omega = H$, $X(\omega) = 1$
 $X: \Omega \rightarrow \{1, 0\}$

A pricing path time = $0, 1, \dots, T$

$$\Omega = (0, \infty)^{T+1}$$

$$\omega = (S_0, S_1, \dots, S_T)$$

one entire realized
market pricing path

$$S_t(\omega) := S_t$$

fixed $t=1$ $S_1(\omega) = S_1$ r.v.

$S(\omega)(t)$: stochastic process.

Events collect outcomes for which a property holds

Definition 2 (Event) A property of outcomes is represented by a subset $A \subseteq \Omega$. Once an admissible event family $\mathcal{F}$ has been fixed, the events in the probability model are exactly the sets $A \in \mathcal{F}$. The event A occurs precisely when the realized outcome satisfies $\omega \in A$.

$$2 \text{ coins } \Omega_{\text{coin}} = \{HH, HT, TH, TT\} = \mathcal{F} = 2^{\{H, T\}}$$

$$A = \{\underline{H}H, H\underline{T}\}, \quad B = \{H\underline{H}, T\underline{H}\}.$$

$$A \subseteq \Omega_{\text{coin}}, \quad B \subseteq \Omega_{\text{coin}}.$$

A : event for 1st = H.

B : event for 2nd = H

$$A^c, \quad A \cap B, \quad A \cup B$$

↓
at least one is head.

Define infinite tosses.

$$\Omega_\infty = \{H, T\}^\mathbb{N}$$

$$H_n = \{\omega \in \Omega_\infty : \omega_n = H\}$$

then we have

or

$$\bigcup_{n \geq 1} H_n = \{\text{at least one } H\}.$$

and

$$\bigcap_{n \geq 1} H_n^c = \{\text{no head ever occurs}\}.$$

$$\bigcap_{N \geq 1} \bigcup_{n \geq N} H_n = \{\text{infinite many heads occur}\}.$$

$\forall N \geq 1 \exists n \geq N$

$$\limsup_{n \rightarrow \infty} H_n$$

$$\left(\bigcap_{N \geq 1} \bigcup_{n \geq N} H_n \right)^c = \bigcup_{N \geq 1} \bigcap_{n \geq N} H_n^c = \{\text{finite many heads occur}\}.$$

$\exists N \forall n \geq N$

$$\liminf_{n \rightarrow \infty} H_n^c$$

finite $\sum_{\omega \in \Omega} p(\omega) = 1$

countable $\sum_{\omega \in \Omega} p(\omega)$

uncountable

measure,

assign the mass?

Probability weights must respect countable decompositions

Normalization and countable additivity On the admissible event family, a probability assignment satisfies $\mathbb{P}(\Omega) = 1$ and, for pairwise disjoint events $A_1, A_2, \dots$,

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n)$$

Two tossing {HH, HT, TH, TT} $\mathbb{P}(\{H, \cdot\}) = \frac{1}{2}$

Finite atomic model $\sum_{\omega \in \Omega} \mathbb{P}(\omega) = 1$

For uniform prob. on $[0,1]$, $\mathbb{P}(\{x\}) = 0$

$$\mathbb{P}([0,1]) = 1, [0,1] = \bigcup_{x \in [0,1]} \{x\}$$

$$1 = \mathbb{P}([0,1]) \neq \sum_{x \in [0,1]} \mathbb{P}(\{x\}) = 0$$

A sigma-algebra is closed under countable event logic

Definition 3 (Sigma-algebra) A family $\mathcal{F} \subseteq 2^\Omega$ is a sigma-algebra if

① $\Omega \in \mathcal{F}$, ② $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$, ③ $(A_n)_{n \geq 1} \subseteq \mathcal{F}$ then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.

example $\Omega = \{HH, HT, TH, TT\}$

$A = \{HH, HT\}$

$\mathcal{F} = \{\emptyset, \Omega, A, A^c\}$ σ -algebra.

The event domain is now precise...

$\mathcal{F} \rightarrow [0, 1]$ assigns weights to admissible events in $\mathcal{F}$.

$\mathcal{F}$ can distinct A, A^c
 2^Ω distinct HH, HT

Why not always choose 2^Ω ?

It is a valid sigma-algebra, but not an automatic choice 2^Ω is always a sigma-algebra. The question is whether every subset represents an event to which the model can assign a meaningful and consistent probability.

$$(\Omega, \mathcal{F} \subseteq 2^\Omega, \mathbb{P})$$

Sometimes it works

For a finite sample space, one usually takes all subsets. Even on $\mathbb{R}$, the point mass

$$\delta_0(E) = \mathbf{1}_{\{0 \in E\}}$$

is a probability measure on every subset $E \subseteq \mathbb{R}$.

Why it may be too large

- The recorded observations may not distinguish every subset of Ω .
- Continuous length cannot be countably additive and translation invariant on every subset of $\mathbb{R}$; the Vitali example will show why.

Vitali set for the continuous length.

$$\Omega = \{HH, HT, TH, TT\}, A = \{HT, HH\}$$

	HH	HT	TH	TT	$P(A)$	$P(HH)$
Law 1	1/2	0	1/2	0	1/2	1/2
Law 2	0	1/2	0	1/2	1/2	0

both laws agree on $\mathcal{F} = \{\emptyset, A, A^c, \Omega\}$

But on $\{HH\} \in 2^\Omega$ Law 1 and Law 2 disagree.

The generated sigma-algebra adds exactly what is forced

Definition 4 (Generated sigma-algebra) For a family $\mathcal{C} \subseteq 2^\Omega$, define

$$\sigma(\mathcal{C}) := \bigcap \{ \mathcal{G} : \mathcal{G} \text{ is a sigma-algebra on } \Omega, \mathcal{C} \subseteq \mathcal{G} \}.$$

The collection is nonempty because 2^Ω is one of its members.

Proposition 5 (Minimality) $\sigma(\mathcal{C})$ is a sigma-algebra containing $\mathcal{C}$, and

$$\mathcal{C} \subseteq \mathcal{G}, \mathcal{G} \text{ is a } \sigma\text{-algebra then } \sigma(\mathcal{C}) \subseteq \mathcal{G}.$$

$$\textcircled{1} \quad \Omega = \{HH, HT, TH, TT\} \quad A = \{HH, HT\}.$$

$$\sigma(A) = \{\emptyset, A, A^c, \Omega\}.$$

$\textcircled{2}$ Borel σ -algebra.

$$\mathcal{B}(\mathbb{R}) = \sigma\{(-\infty, x] : x \in \mathbb{R}\}.$$

the event $\{\omega \in \Omega : X(\omega) \leq x\}$ is admissible.

$$\text{CDF} \quad F_X(x) = \mathbb{P}(X \leq x)$$

$$(\Omega, \mathcal{F}, \mathbb{P})$$

$$\{\omega \in \Omega : a < X(\omega) \leq b\}$$

$$= \{\omega \in \Omega : X(\omega) \leq b\}$$

$$- \{\omega \in \Omega : X(\omega) \leq a\},$$

Measure spaces separate the event domain from the weights

Definition 6 (Measure space)

A measure space is a triple $(S, \mathcal{M}, \mu)$, where $\mathcal{M}$ is the σ -algebra on S and $\mu: \mathcal{M} \rightarrow [0, \infty]$ satisfies for every pairwise disjoint seq. $(A_n)_{n \geq 1} \subseteq \mathcal{M}$ it holds $\mu(\emptyset) = 0$, $\mu(\bigcup_{n \geq 1} A_n) = \sum_{n \geq 1} \mu(A_n)$.

Definition 7 (Probability space)

$(\Omega, \mathcal{F}, \mathbb{P})$ is a measure space

① $\mathbb{P}(\Omega) = 1$ ② $\mathbb{P}: \mathcal{F} \rightarrow [0, 1]$ ③ $\mathbb{P}(\bigcup_{n \geq 1} A_n) = \sum_{n \geq 1} \mathbb{P}(A_n)$.

$\Omega = \mathbb{R}$, $\mathcal{B}(\mathbb{R}) = \mathcal{F}$, $\mathbb{P}: \mathcal{B}(\mathbb{R}) \rightarrow [0, 1]$. $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathbb{P})$.

Continuity from below

$$(\mathcal{S}, \mathcal{M}, \mu), \quad \mu: \mathcal{M} \rightarrow [0, \infty]$$

Theorem 8 If $E_n \in \mathcal{S}$ and $E_n \nearrow E := \bigcup_n E_n$, then $\mu(E_n) \rightarrow \mu(E)$.

proof. $D_1 := E_1$

$$D_n := E_n \setminus E_{n-1} \quad \text{for } n \geq 2$$

$\{D_n\}_{n \geq 1}$ are disjoint.

then

$$\begin{aligned} \mu(E_n) &= \mu\left(\bigcup_{n=1}^N D_n\right) = \sum_{n=1}^N \mu(D_n) \rightarrow \sum_{n=1}^{\infty} \mu(D_n) \\ &= \mu\left(\bigcup_{n=1}^{\infty} D_n\right) \\ &= \mu(E). \end{aligned}$$

#

Continuity from above and its necessary hypothesis

$$E_1 \cap E_2 \cap$$



Theorem 9 If $E_n \in \mathcal{S}$, $E_n \searrow E := \bigcap_n E_n$, and $\mu(E_1) < \infty$, then $\mu(E_n) \rightarrow \mu(E)$.

Pf. $E_1 \setminus E_n \rightarrow E_1 \setminus E$, then $\mu(E_1 \setminus E_n) \rightarrow \mu(E_1 \setminus E)$

since $\mu(E_1) < \infty$, we get

$$\mu(E_n) = \mu(E_1) - \mu(E_1 \setminus E_n) \quad (\infty - \infty?)$$

$$\xrightarrow{n \rightarrow \infty} \mu(E_1) - \mu(E_1 \setminus E)$$

$$= \mu(E). \quad \#.$$

Example. Without $\mu(E_1) < \infty$,

$$E_n = [n, \infty) \quad E_n \rightarrow \emptyset \text{ as } n \rightarrow \infty$$

$$\mu(E_n) = \infty \text{ for every } n$$

$$\lim_{n \rightarrow \infty} \mu(E_n) = \infty \neq \mu(\emptyset) \quad \#$$

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Algebra $\rightarrow$ pre-measure $\rightarrow$ outer-measure $\rightarrow$ extension
then
 $\rightarrow$ Lebesgue measure.

Jordan measure $\rightarrow$ outer-measure $\rightarrow$ extension
 $\rightarrow$ Lebesgue measure.

Jordan content uses finite inner and outer approximation

Elementary sets are $\emptyset$ and finite disjoint unions of bounded intervals; their content is the sum of component lengths.

Definition 10 (Jordan inner and outer content) For bounded $E \subseteq \mathbb{R}$,

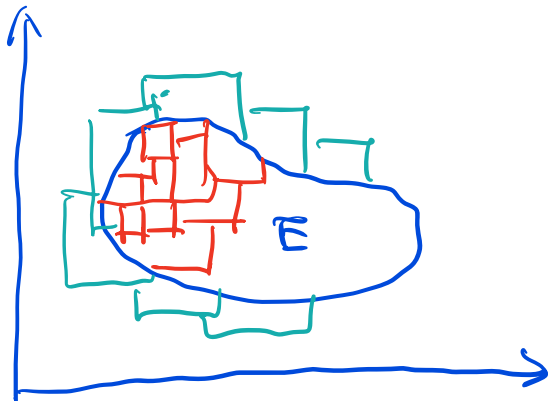
$$J_*(E) := \sup\{|K| : K \subseteq E, K \text{ elementary}\}, \quad E = \bigcup_{k=1}^N R_k$$

$$J^*(E) := \inf\{|O| : E \subseteq O, O \text{ elementary}\}. \quad R_k = [a_k, b_k]$$

E is Jordan measurable when $J_*(E) = J^*(E)$; the common value is $J(E)$.

Theorem 11 (Jordan–Riemann correspondence) For $E \subseteq [a, b]$, E is Jordan measurable if and only if $\mathbf{1}_E$ is Riemann integrable. In that case,

$$J(E) = \int_a^b \mathbf{1}_E(x) dx.$$



Jordan measure built on

finite interval approximation.

$$J_*(E) = \sup \left\{ \left| \bigcup_{k=1}^n R_k \right|, \bigcup_{k=1}^n R_k \subseteq E \right\}.$$

$$J^*(E) = \inf \left\{ \left| \bigcup_{k=1}^n R_k \right|, \bigcup_{k=1}^n R_k \supseteq E \right\}.$$

$$\mu_J^*(\cup A_k) \leq \sum \mu_J^*(A_k)$$

A_k sets.

$$\mu_J^*(E + \tau) = \mu_J^*(E)$$

$E \subseteq \mathbb{R}, \tau \in \mathbb{R}.$

Finite covers see density; countable covers follow points

Enumerate $D = \{q_1, q_2, \dots\}$ $D = \mathbb{Q} \cap [0, 1]$ $\underline{J_*(D)} = 0$, $\underline{J^*(D)} = 1$.

p.f. Every elementary $K \subseteq D$, $|K| = 0$. $J_*(D) = 0$.

Now consider an elementary O covers D , then finite interval structure and density gives

$$[0, 1] = \overline{D} \subseteq \overline{O} \text{ closure, } 1 \leq |\overline{O}| = |O|$$

$$J^*(D) \geq 1, \quad D \subseteq [0, 1], \quad J^*(D) = 1.$$

#



$$q_n \in I_n \quad |I_n| < \varepsilon \cdot 2^{-n}$$

$$D \subseteq \bigcup I_n$$

$$\sum |I_n| < \varepsilon.$$

Outer measure is the infimum of countable covering costs

Definition 12 (Lebesgue outer measure) For every $E \subseteq \mathbb{R}$,

→ Countable sets In

$$\mu^*(E) = \inf \left\{ \sum_{n=1}^{\infty} |I_n| : E \subseteq \bigcup_{n=1}^{\infty} I_n \right\}$$

where I_n range over bounded intervals.

$$A \overset{I_n}{\underbrace{\quad \quad}} \xrightarrow{\text{pool}} A \cup B \subseteq (\cup I_n) \cup (\cup J_n)$$

$$B \quad \text{---} \quad \text{---}$$

J_n

$$\mu^*(A \cup B) \leq \mu^*(A) + \mu^*(B).$$

Outer measure is not yet a measure on $2^{\mathbb{R}}$

Outer measure is not yet a measure on $2^{\mathbb{R}}$ The value $m^*(E)$ exists for every subset E , but only monotonicity and countable subadditivity are guaranteed on the full power set. Even for disjoint sets, equality need not hold. A later measurability criterion will select the domain on which m^* becomes countably additive.

K cover AUB

split? $\rightarrow$

X

$K \cap A$ - - - - -

$K \cap B$ - - - - -

fragmented pieces?

cannot infer $\rightarrow \mu^*(A \cup B) \geq \mu^*(A) + \mu^*(B)$

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The Carathéodory test asks whether every cut preserves cost

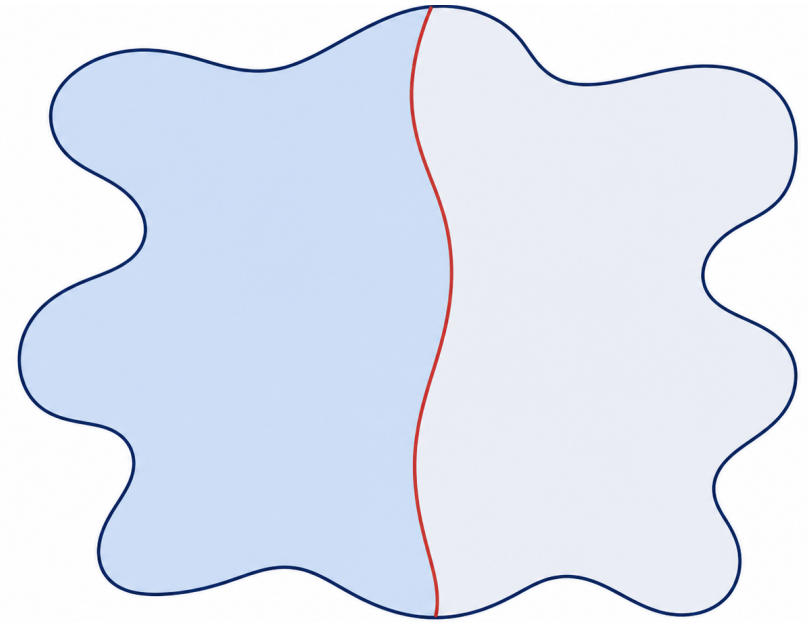
Definition 13 (Carathéodory measurable set)

For an outer measure m^* , a set $E \subseteq \mathbb{R}$ passes the Carathéodory test if, for every $A \subseteq \mathbb{R}$,

$$m^*(A) \geq m^*(A \cap E) + m^*(A \setminus E). \quad (C)$$

Theorem 14 (Carathéodory measurable set theorem)

The sets satisfying (C) form a sigma-algebra, and m^* restricted to this sigma-algebra is a complete measure. For Lebesgue outer measure, (C) is equivalent to the open approximation criterion.



$$A = \underbrace{(A \cap E)} + \underbrace{(A \setminus E)}.$$

A is cut by E creates no extra cost
covering.

$(S, \mathcal{M}, \mu)$
↑

Jordan → outer → Carathéodory.

Vitali theorem: length cannot live on every subset

Theorem 15 (Vitali impossibility theorem) Assuming the axiom of choice, there is no countably additive measure

$$\mu : 2^{\mathbb{R}} \longrightarrow [0, \infty]$$

such that, for every $E \subseteq \mathbb{R}$, $t \in \mathbb{R}$, and $a < b$,

$$\mu(E + t) = \mu(E), \quad \mu([a, b)) = b - a.$$

Construct a Vitali selector On $[0, 1)$, declare

$$x \sim y \iff x - y \in \mathbb{Q}.$$

Using choice, select exactly one representative from each equivalence class and call the resulting set V . Put

$$Q_0 := \mathbb{Q} \cap (-1, 1), \quad U := \bigcup_{q \in Q_0} (V + q).$$

Why the translates work

- If $v_1 + q_1 = v_2 + q_2$, then $v_1 - v_2 = q_2 - q_1 \in \mathbb{Q}$. Hence $v_1 \sim v_2$, so uniqueness of representatives gives $v_1 = v_2$ and $q_1 = q_2$. Thus the translates are pairwise disjoint.
- For $x \in [0, 1)$, its representative $v \in V$ gives $q = x - v \in Q_0$, so $x \in V + q$. Since $V \subseteq [0, 1)$ and $q \in (-1, 1)$,

$$[0, 1) \subseteq U \subseteq [-1, 2).$$

Countable additivity forces the contradiction

Proof Assume that such a measure μ exists and set $d := \mu(V)$. Since $V \subseteq [0, 1)$, monotonicity gives $0 \leq d \leq 1$. Translation invariance, pairwise disjointness, and countable additivity give

$$\mu(U) = \sum_{q \in Q_0} \mu(V + q) = \sum_{q \in Q_0} d.$$

But the interval sandwich gives

$$1 = \mu([0, 1)) \leq \mu(U) \leq \mu([-1, 2)) = 3.$$

- If $d = 0$, then $\mu(U) = 0$, contradicting $\mu(U) \geq 1$.
- If $d > 0$, then $\mu(U) = \infty$, contradicting $\mu(U) \leq 3$.

Thus no such measure on $2^{\mathbb{R}}$ exists; in particular, V is not Lebesgue measurable.

The construction delivers exactly what probability needs

Theorem 16 (Structure of Lebesgue measure) The Lebesgue measurable sets and Lebesgue measure satisfy

$$\mathcal{B}(\mathbb{R}) \subseteq \mathcal{L}(\mathbb{R}) \subsetneq 2^{\mathbb{R}}, \quad m([a, b]) = b - a \quad (a \leq b),$$

and, for pairwise disjoint $E_n \in \mathcal{L}(\mathbb{R})$,

$$m\left(\bigcup_{n \geq 1} E_n\right) = \sum_{n \geq 1} m(E_n).$$

A measurable set Z with $m(Z) = 0$ is called null. Lebesgue measure is complete: every subset $N \subseteq Z$ is measurable and null.

Corollary 17 (Uniform probability) Let $\mathcal{L}([0, 1]) := \{E \cap [0, 1] : E \in \mathcal{L}(\mathbb{R})\}$. Then

$$\mathbb{P}(A) := m(A), \quad A \in \mathcal{L}([0, 1]),$$

defines the uniform probability measure on $[0, 1]$.

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Darboux sums make the Riemann criterion explicit

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded and let

$$P : \quad a = x_0 < x_1 < \cdots < x_n = b, \quad I_j = [x_{j-1}, x_j]$$

For each interval cell, put

$$m_j := \inf_{x \in I_j} f(x), \quad M_j := \sup_{x \in I_j} f(x).$$

Lower and upper Darboux sums

$$L(f, P) = \sum_{j=1}^n m_j |I_j| \quad U(f, P) = \sum_{j=1}^n M_j |I_j|$$

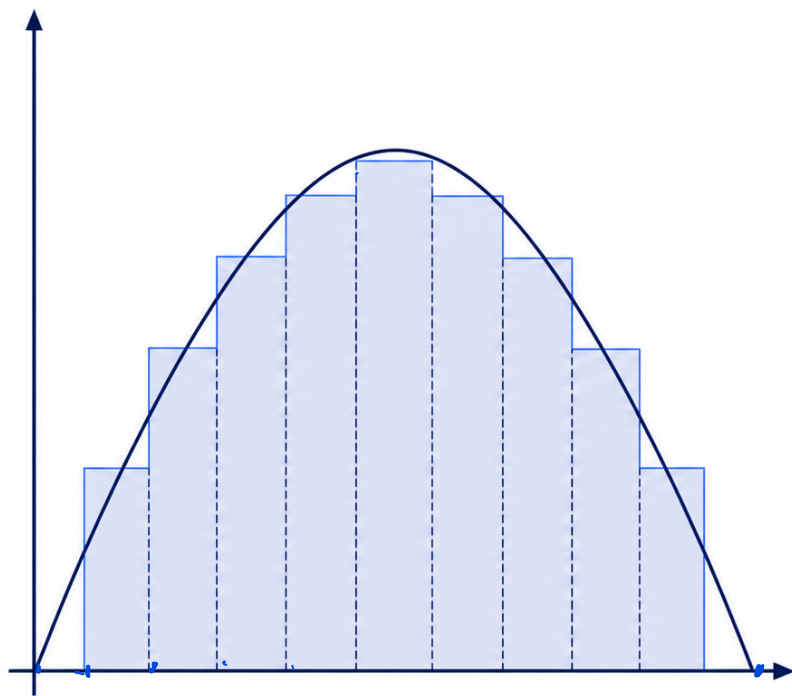
f is Riemann integrable

$$\Leftrightarrow \forall \varepsilon > 0, \exists P : U(f, P) - L(f, P) < \varepsilon$$

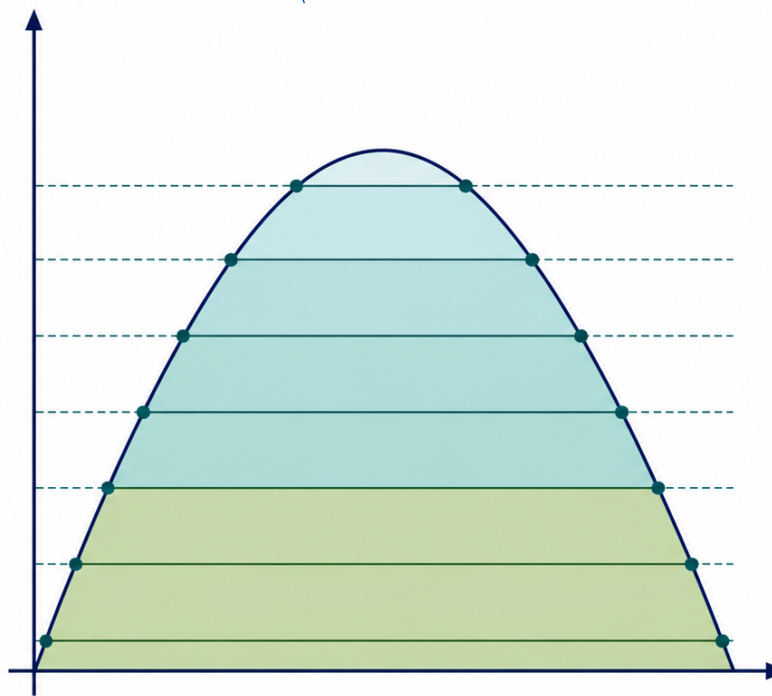
Fixed ε . we ignore the oscillation in I_i
the oscillation on total is small.

Lebesgue keeps the staircases but changes their base sets

Darboux: interval cells only



Lebesgue: measurable level sets



What changes

$$\mathbb{1}_{\Sigma \cap \mathbb{R} \cap \mathbb{Q}}$$

Riemann-Darboux: the lower and upper staircases are tied to finitely many interval cells. Refining P must force $U(f, P) - L(f, P)$ to zero.

Lebesgue: a simple staircase $s = \sum_k a_k \mathbb{1}_{A_k}$ may use arbitrary measurable sets A_k . Taking all measurable simple sets can isolate scattered or dense null behavior instead of charging the surrounding intervals.

Lebesgue integral permits discontinuous function.
 $\Leftrightarrow$ discontinuous set is a null set. $du(N)=0$

Bounded Riemann integrability implies Lebesgue integrability

Theorem 18 (Riemann–Lebesgue compatibility) Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. If f is properly Riemann integrable, then f is Lebesgue measurable and integrable, and

$$\int_a^b f(x) dx \Big|_{\mathbb{R}} = \int_{[a,b]} f dm \Big|_{\mathbb{L}}.$$

Why measurability and integrability hold The Darboux criterion implies that the discontinuity set of f is Lebesgue null: cells with oscillation at least η have total length at most $(U - L)/\eta$. Hence f is Lebesgue measurable. If $|f| \leq M$, then

$$\int_{[a,b]} |f| dm \leq M(b - a) < \infty.$$

Why the two values agree For each partition P , its lower and upper Darboux step functions satisfy

$$\ell_P \leq f \leq u_P, \quad \int \ell_P dm = L(f, P), \quad \int u_P dm = U(f, P).$$

Both integral values are trapped in $[L(f, P), U(f, P)]$. Since the Darboux gap can be made smaller than every $\varepsilon > 0$, the two values are equal.

The next slides construct the simple-function integral that makes the displayed Darboux-step identities formal.

Lebesgue integrability does not imply Riemann integrability

Counterexample 19 (Dirichlet function) Let

$$f = \mathbf{1}_{\mathbb{Q} \cap [0,1]}.$$

$$\textcircled{1} \quad U(f, P) = 1 \quad L(f, P) = 0, \quad U(f, P) - L(f, P) = 1$$

This is not Riemann-integrable.

$\textcircled{2}$ $\mathbb{Q} \cap [0,1]$ is countable, it is Lebesgue measure
and $\{q_1, q_2, \dots\}$, $\text{Leb}(\{q_i\}) = 0$ null set!

$$\int_{[0,1]} f \, d\text{Leb} = \text{Leb}([0,1] \cap \mathbb{Q}) = 0.$$

Lebesgue measure.

Measurable functions are the inputs to the integral

Definition 20 (Extended-real measurable function) Let $(S, \underline{\mathcal{S}})$ be a measurable space. A map

$$f: S \rightarrow [-\infty, \infty]$$

is measurable when

is measurable.

$$\{x : f(x) > a\} \in \mathcal{S} \quad \text{for every } a \in \mathbb{R}.$$

Equivalent tests use $\{f \geq a\}$, $\{f < a\}$, or $\{f \leq a\}$.

Integrable: $\int |f| d\mu < \infty$

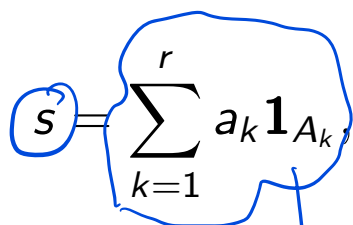
$1/x$ measure but not integral on $[0,1]$.

Indicators are the atomic measurable functions

Definition 21 (Indicator)

$$\mathbf{1}_A(x) := \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases} \quad (A \in \mathcal{S}).$$

Definition 22 (Nonnegative simple function) A measurable function $s : S \rightarrow [0, \infty)$ is a nonnegative simple function if its range is finite. If its distinct values are $a_1, \dots, a_r$, then its canonical representation is



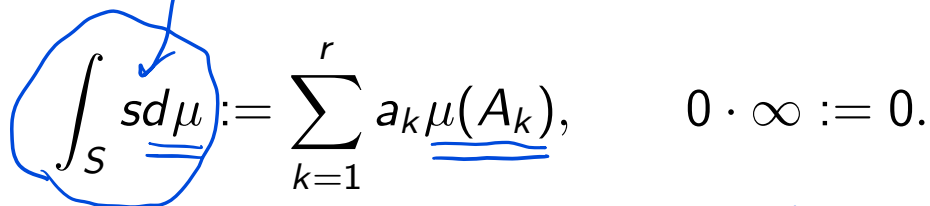
A handwritten blue circle around the variable s in the equation $s = \sum_{k=1}^r a_k \mathbf{1}_{A_k}$. A blue arrow points from this circle down to the integral in Definition 23.

$$s = \sum_{k=1}^r a_k \mathbf{1}_{A_k}$$

classify by the range

$$A_k = \{s = a_k\}, \quad S = \bigcup_{k=1}^r A_k.$$

Definition 23 (Simple integral)



A handwritten blue circle around the integral $\int_S s d\mu$ in the equation $\int_S s d\mu := \sum_{k=1}^r a_k \mu(A_k)$. A blue arrow points from this circle up to the variable s in the equation in Definition 22.

$$\int_S s d\mu := \sum_{k=1}^r a_k \mu(A_k), \quad 0 \cdot \infty := 0.$$

$$\mathbf{1}_A = 1 \cdot \mathbf{1}_A + 0 \cdot \mathbf{1}_{A^c}$$

Corollary 24 (Indicator identity) For $A \in \mathcal{S}$, $\int_S \mathbf{1}_A d\mu = \mu(A)$.

$$= \mu(A) + 0 = \mu(A). \quad \#$$

The simple integral is representation independent

Proposition 25 If

$$s = \sum_{i=1}^p a_i \mathbf{1}_{A_i} = \sum_{j=1}^q b_j \mathbf{1}_{B_j},$$

where each displayed family is a measurable partition of S , then the two formulas for $\int s d\mu$ agree.

proof. We consider $A_i \cap B_j \neq \emptyset$, we have $a_i = b_j$.

$$\int s d\mu = \sum_i a_i \mu(A_i) = \sum_i a_i \sum_j \mu(A_i \cap B_j)$$

$$= \sum_{i,j} b_j \mu(A_i \cap B_j)$$

$$= \sum_j b_j \mu(B_j).$$

#.

Linearity and monotonicity for simple functions

Proposition 26 For nonnegative simple s, t and $\alpha, \beta \geq 0$,

$$\int (\alpha s + \beta t) d\mu = \alpha \int s d\mu + \beta \int t d\mu.$$

If $s \leq t$, then $\int s d\mu \leq \int t d\mu$.

$$s = \sum a_i 1_{A_i} \quad t = \sum b_j 1_{B_j}.$$

Simple $\rightarrow$ Lebesgue integral, how about general
function f . ?

Extend the simple integral by lower approximation

We know how to integrate nonnegative simple functions. For a general measurable $f \geq 0$ use every measurable simple staircase lying below it.

Definition 27 (Lebesgue integral for $f \geq 0$) For measurable $f : S \rightarrow [0, \infty]$, let

$$\mathcal{S}_f := \{s : 0 \leq s \leq f, \text{ } s \text{ simple and measurable}\}, \quad \int_S f d\mu := \sup_{s \in \mathcal{S}_f} \int_S s d\mu.$$

The supremum need not be attained, and the value $+\infty$ is allowed.

Proposition 28 (Immediate results) If f is nonnegative simple, this definition agrees with its simple integral. If $0 \leq f \leq g$ and $c \geq 0$, then

$$\int f d\mu \leq \int g d\mu, \quad \int c f d\mu = c \int f d\mu.$$

$$f \in \mathbb{R}$$

$$f^+ = f \vee 0$$



Lebesgue

$$f^- = (-f) \vee 0$$



Lebesgue integral

$$f = f^+ - f^-$$

Measurability alone gives an increasing simple approximation

Considering increasing sequence of functions.

Theorem 29 If $f : S \rightarrow [0, \infty]$ is measurable, there are nonnegative measurable simple functions s_n such that $s_n \uparrow f$ pointwise.

Canonical dyadic approximation Define

$$s_n := 2^{-n} \sum_{k=1}^{n2^n} \mathbf{1}_{\{f \geq k2^{-n}\}}.$$

At each point, $\underline{s_n}$ is $\underline{f \wedge n}$ rounded down to the nearest multiple of 2^{-n} .

We check:

1. $\{f \geq k2^{-n}\}$ is measurable, sum is finite, s_n is simple and measurable.

2. Increasing gives $0 \leq s_n \leq s_{n+1} \leq f$

3. If $f(x) < \infty$, we have $0 < f(x) - s_n(x) < 2^{-n}$.

If $f(x) = \infty$, then $s_n(x) = n$.

Hence, $s_n \uparrow f$ p.w. For every $x \in \mathbb{R}$, can we have

$$\int s_n(x) d\mu \uparrow \int f(x) d\mu ?$$

Where we are

1. Probability spaces and measurable events
2. From Jordan content to outer measure
3. Lebesgue measurable sets and measure
4. Lebesgue integration
5. MCT, Fatou's Lemma, DCT, BCT

Monotone convergence theorem

Theorem 30 (Monotone convergence theorem)

Let $(E, \mathcal{M}, \mu)$ be any measure space. If $f_n: M \rightarrow [0, \infty]$ are measurable and $f_n(x) \uparrow f(x)$ for every $x \in M$ then f is measurable and $\int f_n d\mu \uparrow \int f d\mu$.

nonnegative
↓

Pf. NTS. $\int f_n \leq \int f$

Since we have $\{f > a\} = \bigcup_n \{f_n > a\}$

Monotonicity gives

$$\int f_n d\mu \leq \int f_{n+1} d\mu \leq \int f d\mu$$

then $\lim_{n \rightarrow \infty} \int f_n d\mu$ exists and $\lim_{n \rightarrow \infty} \int f_n d\mu \leq \int f d\mu$.

$$\underline{NTS} \quad \int f_n \geq \int f$$

$$\int f_n \geq (1-\varepsilon) \int f \quad \forall \varepsilon > 0$$

$\uparrow$
 $\varepsilon \rightarrow 0$
 epsilon room.

For a nonnegative simple $0 \leq s \leq f$

we take $c \in (0,1)$. Let

$$E_n := \{x : f_n(x) \geq c s(x)\} \quad (*)$$

E_n is measurable because $f_n - cs$ is measurable.

and $E_n \subseteq E_{n+1}$ because $f_n \leq f_{n+1}$. Then we have

$$s(x) = 0 \Rightarrow f_n(x) > 0 = cs(x)$$

$$s(x) > 0 : f_n(x) \uparrow f(x) \geq s(x) > cs(x)$$

$$E_n \uparrow E := \{x : f(x) \geq cs(x)\}.$$

$$E = \bigcup_{j=1}^r A_j$$

Now if $s = \sum_{j=1}^r a_j 1_{A_j}$ then $A_j \cap E_n \uparrow A_j \cap E = A_j$ (**)

$$\int s 1_{E_n} d\mu = \sum_{j=1}^r a_j \mu(A_j \cap E_n) \uparrow \sum_{j=1}^r a_j \mu(A_j) = \int s d\mu.$$

$$\text{so } I := \lim_{n \rightarrow \infty} \int f_n d\mu, \quad I \geq c \int s d\mu$$

$$\text{because } f_n \geq cs 1_{E_n} \quad (**).$$

If $\int s = \infty$, then $I = \infty$ for $c > 0$.

If $\int s < \infty$, let $c \uparrow 1$, $0 \leq s \leq f$

$$I \geq \int s d\mu \Rightarrow I \geq \sup_{f \geq s} \int s d\mu = \int f d\mu. \quad \#$$

MCT extends simple-function additivity to all $f, g \geq 0$

Proposition 31 For measurable $f, g : S \rightarrow [0, \infty]$,

$$\int (f + g) d\mu = \int f d\mu + \int g d\mu.$$

Pf. $s_n \uparrow f$, $t_n \uparrow g$, $s_n + t_n \uparrow f + g$, s_n, t_n are simple

$$\int (f + g) d\mu = \int \lim_{n \rightarrow \infty} (s_n + t_n) d\mu$$

$$= \lim_{n \rightarrow \infty} \left(\int s_n d\mu + \int t_n d\mu \right)$$

$$= \int f d\mu + \int g d\mu \quad \text{MCT.}$$

#

Pointwise convergence alone does not preserve mass

If there is no monotonic property, can we still hold the statement?

$$f_n \rightarrow f \not\Rightarrow \int f_n \rightarrow \int f$$

The measure space $([0,1], \mathcal{B}[0,1], \text{Leb})$, let

$$f_n(x) = n \mathbf{1}_{[0, 1/n]}(x)$$

$$f_n \rightarrow f = 0 \quad n \rightarrow \infty \text{ p.w.}$$

$$\int_{[0,1]} f_n \, d\text{Leb} = \int_{[0,1]} n \mathbf{1}_{[0, 1/n]}(x) \, d\text{Leb} = n \cdot \text{Leb}(0, 1/n) = 1$$

$$\lim_{n \rightarrow \infty} \int_{[0,1]} f_n \, d\text{Leb} = 1 \neq 0 = \int_{[0,1]} \overset{\lim_{n \rightarrow \infty} f_n}{f} \, d\text{Leb}$$

#

Tail infima recover the monotonicity needed by MCT

classical.

Definition 32 (Pointwise lower and upper limits) For extended real numbers a_n ,

$$\liminf_{n \rightarrow \infty} a_n := \sup_{N \geq 1} \inf_{n \geq N} a_n, \quad \limsup_{n \rightarrow \infty} a_n := \inf_{N \geq 1} \sup_{n \geq N} a_n.$$

In fact, if we define $g_N := \inf_{n \geq N} a_n$ $\uparrow$ $\liminf_{n \rightarrow \infty} a_n$

$$h_N := \sup_{n \geq N} a_n \downarrow \limsup_{n \rightarrow \infty} a_n = \inf_{N \geq 1} \sup_{n \geq N} a_n.$$

$\{A_n\}_{n=1}^{\infty}$ are a sequence of sets. Define $\liminf_{n \rightarrow \infty} A_n$,

$$\limsup_{n \rightarrow \infty} A_n.$$

$$\limsup_{n \rightarrow \infty} A_n = \bigcap_{N \geq 1} \bigcup_{n \geq N} A_n$$

$$\liminf_{n \rightarrow \infty} A_n = \bigcup_{N \geq 1} \bigcap_{n \geq N} A_n$$

$$(\limsup A_n)^c = (\bigcap \bigcup A_n)^c = \bigcup \bigcap A_n^c$$

Fatou's lemma

→ nonnegative

Lemma 33 (Fatou) If $f_n : S \rightarrow [0, \infty]$ are measurable on any measure space,

$$\int_S \liminf_{n \rightarrow \infty} f_n d\mu \leq \liminf_{n \rightarrow \infty} \int_S f_n d\mu$$

Pf. Let $g_N = \inf_{n \geq N} f_n$, $\{g_N < a\} = \bigcup_{n \geq N} \{f_n < a\}$

$0 \leq g_N \uparrow \liminf_{n \rightarrow \infty} f_n$ and $\underline{g_N} < \underline{f_n}$ for $n \geq N$.

then we get

by

$$\int \liminf_n f_n = \int \lim_N g_N = \lim_N \int \underline{g_N} \leq \liminf_N \underbrace{\int f_j}_{\#}$$

Reverse Fatou's

$\exists g$ s.t. $0 \leq f_n \leq g$, $\int |g| d\mu < \infty$

$\limsup(-f)$

$$\int \limsup_n f_n \geq \limsup_n \int f_n$$

$$= - \liminf f$$

$$\int \liminf_n (\underline{g - f_n}) \leq \liminf \int (g - f_n) \quad \text{Fatou's} \quad \text{integrable, } \#$$

Example. Let $f(x)$ be nondecreasing on bdd $[0,1]$.

Assuming f is differentiable almost everywhere.
prove that $\int_0^1 f'(x) dx \leq f(1) - f(0)$.

Let $f' = 0$ on the non-differentiable set. null set

$f' \geq 0$ on the differentiable set.

We define

$$g_n(x) = \begin{cases} \frac{f(x + \frac{1}{n}) - f(x)}{1/n} & 0 \leq x \leq 1 - \frac{1}{n} \\ 0 & 1 - \frac{1}{n} \leq x \leq 1 \end{cases}$$

$g_n \rightarrow f'$ from right. $\int_{1/n}^1 + \int_0^{1/n} = \int_0^{1-1/n} + \int_{1-1/n}^1$

Fatou's lemma gives

$$\int_0^1 f'(x) dx \leq \int_0^1 \liminf g_n \leq \liminf \int_0^1 g_n$$

We compute $= n \left(\int_{1/n}^1 f(u) du - \int_0^{1-1/n} f(u) du \right)$.

$$\int_0^1 g_n(x) dx = n \int_0^{1-1/n} [f(x + \frac{1}{n}) - f(x)] dx$$

$$= n \left(\int_{1-1/n}^1 f(u) du - \int_0^{1/n} f(u) du \right)$$

$$\int_{1-1/n}^1 f(u) du \leq \frac{1}{n} f(1) \quad \int_0^{1/n} f(u) du \geq \frac{1}{n} f(0)$$

$$\int_0^1 g_n \leq \left(\frac{1}{n} \right) (f(1) - f(0))$$

$$\int_0^1 f'(x) dx \leq f(1) - f(0).$$

#

On L^1 , null changes and linear combinations are harmless

A statement holds μ -almost everywhere (a.e.) if its failure set is contained in a measurable set N with $\mu(N) = 0$.

Lemma 34 (Null-support principle) If $h \geq 0$ is measurable and $h = 0$ a.e., then $\int h d\mu = 0$.
Indeed, every simple element of h is supported on a null set.

Proposition 35 (L^1 is a stable linear domain) If $f, g \in L^1(\mu)$ and $\alpha, \beta \in \mathbb{R}$, then

$$\alpha f + \beta g \in L^1(\mu), \quad \int (\alpha f + \beta g) d\mu = \alpha \int f d\mu + \beta \int g d\mu,$$

and

$$\left| \int f d\mu \right| \leq \int |f| d\mu.$$

The estimate $|\alpha f + \beta g| \leq |\alpha||f| + |\beta||g|$ gives integrability; nonnegative additivity applied to positive and negative parts gives linearity. Hence $f = g$ a.e. implies $\int f = \int g$.

It means...

$$f_n \rightarrow f \quad \underline{\text{condition}} \quad \int f_n \rightarrow \int f.$$

$$\int (f_n - f) d\mu \rightarrow 0 \Leftrightarrow \left| \int (f_n - f) d\mu \right| \rightarrow 0$$

$$\leftarrow \left| \int (f_n - f) \right| \leq \int |f_n - f| \rightarrow 0 \quad \textcircled{g_n} \geq f_n - f$$

it is the basic idea for dominated convergence thm.

Dominated convergence theorem

Theorem 36 (Dominated convergence theorem)

Let $f_n, f: S \rightarrow \mathbb{R}$ be measurable. Suppose

$f_n \rightarrow f$ μ -a.e., suppose there $\exists$ measurable function

$g: S \rightarrow [0, \infty)$, nonnegative, $|f_n| \leq g$, if $g \in L^1(\mu)$

then $f \in L^1(\mu)$ and

$$\int |g| d\mu < \infty$$

$$\int |f_n - f| d\mu \rightarrow 0, \quad \int f_n d\mu \rightarrow \int f d\mu.$$

pf. We know $f_n \rightarrow f$ μ -a.e. and $|f_n| \leq g \in L^1(\mu)$.

so $|f| \leq g$, $f \in L^1(\mu)$.

$$\text{Then } |f_n - f| \leq |f_n| + |f| \leq 2g.$$

$$\begin{aligned} \int \liminf_n (2g - |f_n - f|) &\leq \liminf_n \int 2g - |f_n - f| \\ &= 2\int g - \limsup_n \int |f_n - f| \\ &\stackrel{2\int g}{=} \end{aligned}$$

$$\Rightarrow \limsup_n \int |f_n - f| d\mu = 0$$

$$\Rightarrow \int |f_n - f| d\mu \rightarrow 0$$

$$\Rightarrow \left| \int f_n - \int f \right| \rightarrow 0$$

$$\Rightarrow \int f_n \rightarrow \int f.$$

#.

Bounded convergence is a finite measure corollary

Corollary 37 (Bounded convergence theorem) Suppose $\mu(S) < \infty$, the functions $f_n, f : S \rightarrow \mathbb{R}$ are measurable, $f_n \rightarrow f$ almost everywhere, ~~there exists a~~ constant M , s.t. $|f_n| \leq M$ a.e. for every n , then

$$\int |f_n - f| d\mu \rightarrow 0, \quad \int f_n d\mu \rightarrow \int f d\mu.$$

Pf. Since $|f_n| \leq M$ a.e. $\forall \delta > 0$,

$$\{|f_n| > M + \delta\} \subseteq \{|f_n - f| > \delta\} \text{ a.e.}$$

$$\mu\{|f| > M + \delta\} \leq \mu\{|f_n - f| > \delta\} \rightarrow 0. \quad f_n \rightarrow f \text{ a.e.}$$

$$B_n \cup G_n = S, \quad B_n = \{|f_n - f| > \varepsilon\} \quad \mu(B_n) \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\int_S |f_n - f| d\mu = \int_{B_n} |f_n - f| d\mu + \int_{G_n} |f_n - f| d\mu$$

$$\leq \varepsilon \mu(B_n) + \varepsilon \mu(G_n)$$

$$\leq \varepsilon \mu(B_n) + \varepsilon \mu(S) \quad \varepsilon \rightarrow 0 \quad \# \\ \rightarrow 0.$$

$(S, \mathcal{M}, \mu)$

→ Lebesgue measure
outer measure



Lebesgue integral

indicator

→ simple → nonnegative

→ signed integrable.



MCT, Fatou,
DCT, BCT.



basis convergence theorem
in measure theory.

$$\text{MCT} : f_n \uparrow f \Rightarrow \int f_n \uparrow \int f$$